Plan

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An Example for Motivation

Solve the simultaneous linear equations:

$$x - y - z = 2, \qquad 3x - 3y + 2z = 16, \qquad 2x - y + z = 9.$$

Step 1: Represent the given equations in a rectangular array form as follows.

$$\begin{array}{rcrcrcr} x & - & y & - & z & = & 2 \\ 3x & - & 3y & + & 2z & = & 16 \\ 2x & - & y & + & z & = & 9 \end{array} \qquad \left[\begin{array}{rrr|r} 1 & -1 & -1 & 2 \\ 3 & -3 & 2 & 16 \\ 2 & -1 & 1 & 9 \end{array}\right]$$

Step 2: Subtract 3 times the 1st equation from the 2nd equation; **Subtract** 3 **times the 1st row from the 2nd row.**

$$\begin{aligned} x - y - z &= 2 \\ 5z &= 10 \\ 2x - y + z &= 9 \end{aligned} \qquad \left[\begin{array}{ccc|c} 1 & -1 & -1 & 2 \\ 0 & 0 & 5 & 10 \\ 2 & -1 & 1 & 9 \end{array}\right]$$

Step 3: Subtract 2 times the 1st equation from the 3rd equation; **Subtract** 2 **times the 1st row from the 3rd row.**

$$\begin{aligned} x - y - z &= 2 \\ 5z &= 10 \\ y + 3z &= 5 \end{aligned} \qquad \left[\begin{array}{ccc|c} 1 & -1 & -1 & 2 \\ 0 & 0 & 5 & 10 \\ 0 & 1 & 3 & 5 \end{array}\right]$$

Step 4: Interchange the 2nd and 3rd equation; **Interchange the 2nd and 3rd row.**

$$\begin{array}{rcrcrcr} x & - & y & - & z & = & 2 \\ & & y & + & 3z & = & 5 \\ & & & & 5z & = & 10 \end{array} \qquad \left[\begin{array}{rrr|r} 1 & -1 & -1 & 2 \\ 0 & 1 & 3 & 5 \\ 0 & 0 & 5 & 10 \end{array}\right]$$

By backward substitution, we find $z = 2, y = -1, x = 3$ is a solution of the given system of equations.

Matrices

Definition:

- A matrix $A = [a_{ij}]$, of **size** $m \times n$, is an array of numbers in m rows and n columns:

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}.$$

- The number a_{ij} is called the (i, j)-th entry of A.
- A $1 \times n$ matrix is called a **row matrix** (or *row vector*) and an $n \times 1$ matrix is called a **column matrix** (or *column vector*).
- Matrices $A = [a_{ij}]$ and $B = [b_{ij}]$ are said to be **equal** if they are of same size and $a_{ij} = b_{ij}$ for each i and j.

- If $m = n$, then A is called a square matrix.
- If A is a square matrix, then the entries a_{ii} are called the diagonal entries of A.
- If A is a square matrix and if $a_{ij} = 0$ for all $i \neq j$, then A is called a diagonal matrix.
- If an $n \times n$ diagonal matrix has all diagonal entries equal to 1, then it is called the identity matrix of size n, and is denoted by I_n (or simply by I).
- If all the entries of A are equal to 0 then A is called a zero matrix, denoted $\mathbf{O}_{m \times n}$ (or simply by $\mathbf{O}$)

- A matrix B is said to be a sub matrix of A if B is obtained by deleting some rows and/or columns of A.
- Transpose A^t of $A = [a_{ij}]$ is defined as $A^t = [a_{ji}]$, for all i, j.
- The matrix A is said to be symmetric if $A^t = A$.
- The matrix A is said to be skew-symmetric if $A^t = -A$.
- If A is a complex matrix, then $\overline{A} = [\overline{a}_{ij}]$ and $A^* = \overline{A}^t$.
- The matrix A^* is called the conjugate transpose of A.
- The (complex) matrix A is said to be Hermitian if $A^* = A$, and skew-Hermitian if $A^* = -A$.

- A square matrix A is said to be upper triangular if $a_{ij} = 0$ for all $i > j$.
- A square matrix A is said to be lower triangular if $a_{ij} = 0$ for all $i < j$.
- Let $A = [a_{ij}]$ and $B = [b_{ij}]$ be two $m \times n$ matrices and $c \in \mathbb{C}$. Then $A + B = [a_{ij} + b_{ij}]$, $A - B = [a_{ij} - b_{ij}]$ and $cA = [ca_{ij}]$.
- Let $A = [a_{ij}]$ and $B = [b_{jk}]$ be two $m \times n$ and $n \times r$ matrices, respectively. Then $AB = [c_{ik}]$ is the $m \times r$ matrix, where

$$c_{ik} = a_{i1}b_{1k} + a_{i2}b_{2k} + \ldots + a_{in}b_{nk}.$$

- Let A be an $n \times n$ square matrix. Then we define $A^0 = I_n, \quad A^1 = A$ and $A^2 = AA$.
- In general, if k is a positive integer, we define the power A^k as follows

$$A^k = \underbrace{AA \dots A}_{k \text{ times}}.$$

If A and $\mathbf{O}$ are matrices of the same size, then $A + \mathbf{O} = A = \mathbf{O} + A, \;\; A - \mathbf{O} = A, \;\; \mathbf{O} - A = -A, \;\; A - A = \mathbf{O}.$

Unless otherwise mentioned, all matrices are taken to have complex numbers as entries.

Method of Induction: [Version I] Let $P(n)$ be a mathematical statement based on all positive integers n. Suppose that $P(1)$ is true. If $k \geq 1$ and if the assumption that $P(k)$ is true gives that $P(k+1)$ is also true, then the statement $P(n)$ is true for all positive integers.

Method of Induction: [Version II] Let i be an integer and let $P(n)$ be a mathematical statement based on all integers n of the set $\{i, i+1, i+2, \ldots\}$. Suppose that $P(i)$ is true. If $k \geq i$ and if the assumption that $P(k)$ is true gives that $P(k+1)$ is also true, then the statement $P(n)$ is true for all integers of the set $\{i, i+1, i+2, \ldots\}$.

Result

Let A be an $m \times n$ matrix, $\mathbf{e}_i$ an $1 \times m$ standard unit row vector, and $\mathbf{e}_j$ an $n \times 1$ standard unit column vector. Then $\mathbf{e}_i A$ is the i-th row of A and $A\mathbf{e}_j$ is the j-th column of A.

Result

Let A be a square matrix and let r and s be non-negative integers. Then $A^r A^s = A^{r+s}$ and $(A^r)^s = A^{rs}$.

Result

Let A, B and C be matrices of size $m \times n$, and let $s, r \in \mathbb{C}$. Then

1. *Commutative Law: $A + B = B + A$.*
2. *Associative Law: $(A + B) + C = A + (B + C)$.*
3. $1A = A$ *and* $s(rA) = (sr)A$.
4. $s(A + B) = sA + sB$ *and* $(s + r)A = sA + rA$.

Result

Let A, B and C be matrices, and let $s \in \mathbb{C}$. Then

1. *Associative Law*: $(AB)C = A(BC)$, *if the respective matrix products are defined.*
2. *Distributive Law*:
   $A(B + C) = AB + AC, \quad (A + B)C = AC + BC,$
   if the respective matrix sum and matrix products are defined.
3. $s(AB) = (sA)B = A(sB)$, *if the respective matrix products are defined.*
4. $I_m A = A = A I_n$, *if A is of size $m \times n$.*

Result

Let A and B be two matrices and $k \in \mathbb{C}$. Then

1. $(A^t)^t = A, \quad (kA)^t = kA^t.$
2. $(A+B)^t = A^t + B^t$ *if A and B are of the same size.*
3. $(AB)^t = B^t A^t$ *if the matrix product AB is defined.*
4. $(A^r)^t = (A^t)^r$ *for any non-negative integer r.*

Partitioned Matrix

- By introducing vertical and horizontal lines into a given matrix, we can partition it into some blocks of smaller sub-matrices.

- For example, three partitions of the matrix A are given below:

$$A = \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 1 & 5 \end{bmatrix}, \quad A = \left[\begin{array}{cc|cc} 1 & 0 & 0 & 2 \\ 0 & 1 & 1 & 1 \\ \hline 0 & 0 & 2 & 1 \\ 0 & 0 & 1 & 5 \end{array}\right],$$

$$A = \left[\begin{array}{c|ccc} 1 & 0 & 0 & 2 \\ 0 & 1 & 1 & 1 \\ \hline 0 & 0 & 2 & 1 \\ 0 & 0 & 1 & 5 \end{array}\right], \quad A = \left[\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 2 & 1 \\ \hline 0 & 0 & 1 & 5 \end{array}\right].$$

- A given matrix can have several partitions possible.
- A block matrix $A = [A_{ij}]$ is a matrix, where each entry A_{ij} is itself a matrix.
- Thus a partition of a given matrix give us a block matrix.
- The first partition of the previous matrix A is the block matrix $\begin{bmatrix} I & B \\ \mathbf{O} & C \end{bmatrix} = \left[\begin{array}{cc|cc} 1 & 0 & 0 & 2 \\ 0 & 1 & 1 & 1 \\ \hline 0 & 0 & 2 & 1 \\ 0 & 0 & 1 & 5 \end{array}\right]$, where $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \mathbf{O} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, B = \begin{bmatrix} 0 & 2 \\ 1 & 1 \end{bmatrix}, C = \begin{bmatrix} 2 & 1 \\ 1 & 5 \end{bmatrix}$.

Result

Let A and B be two matrices of sizes $m \times n$ and $n \times r$, resp.

1. *If* $B = [\mathbf{b}_1 \mid \mathbf{b}_2 \mid \ldots \mid \mathbf{b}_r]$ *then* $AB = [A\mathbf{b}_1 \mid A\mathbf{b}_2 \mid \ldots \mid A\mathbf{b}_r]$.

2. *If* $A = \begin{bmatrix} \mathbf{a}_1 \\ \hline \mathbf{a}_2 \\ \hline \vdots \\ \hline \mathbf{a}_m \end{bmatrix}$ *then* $AB = \begin{bmatrix} \mathbf{a}_1 B \\ \hline \mathbf{a}_2 B \\ \hline \vdots \\ \hline \mathbf{a}_m B \end{bmatrix}$.

3. *If* $m = n = r$ *and if* $A = [\mathbf{a}_1 \mid \mathbf{a}_2 \mid \ldots \mid \mathbf{a}_n]$ *and* $B = \begin{bmatrix} \mathbf{b}_1 \\ \hline \mathbf{b}_2 \\ \hline \vdots \\ \hline \mathbf{b}_n \end{bmatrix}$

   then
$$AB = \mathbf{a}_1\mathbf{b}_1 + \mathbf{a}_2\mathbf{b}_2 + \ldots + \mathbf{a}_n\mathbf{b}_n.$$